

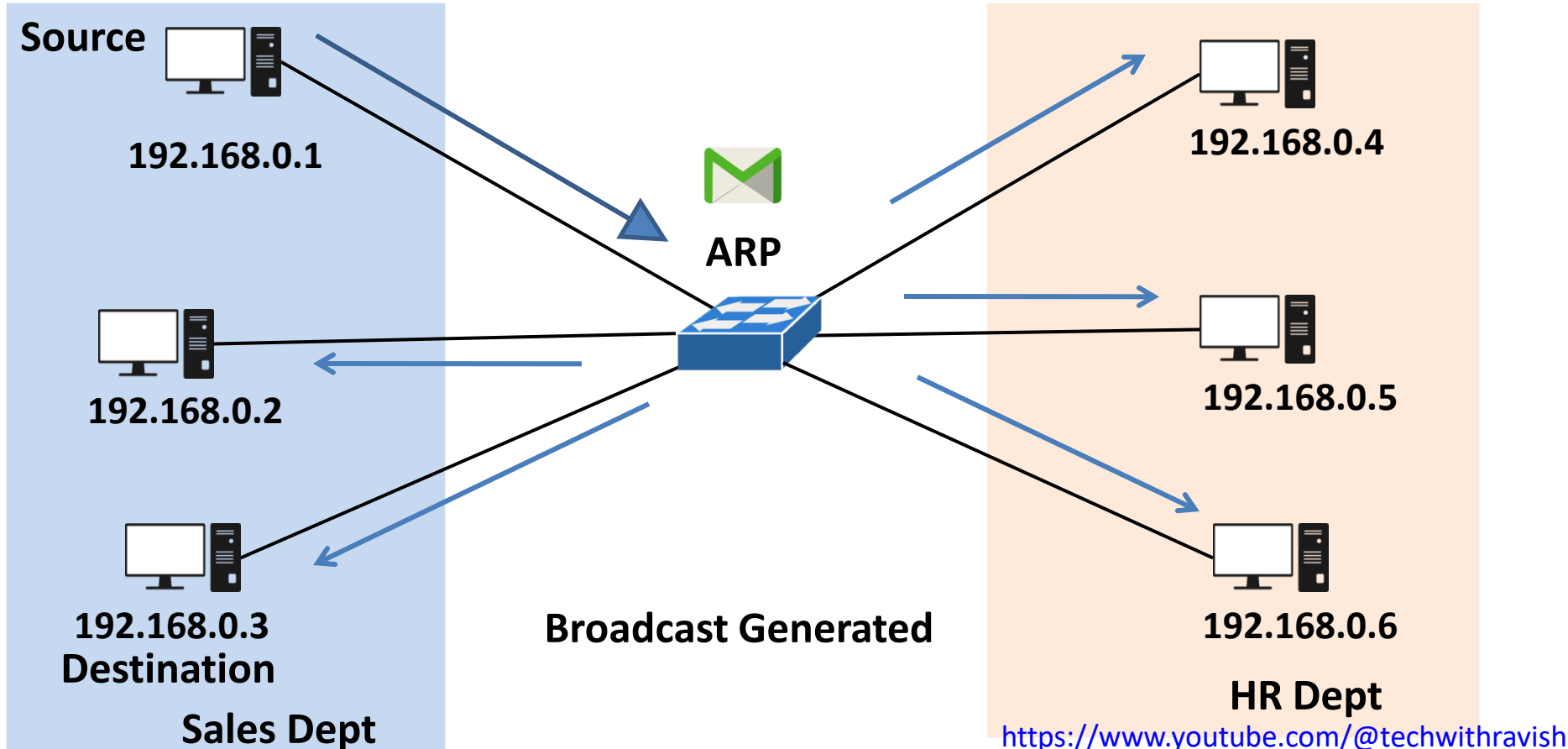
VLAN: Virtual Local Area Network

A VLAN, or Virtual Local Area Network, is a network configuration that allows you to create separate, isolated networks within a single physical network infrastructure. VLANs enable the segmentation of a network into distinct broadcast domains, which can enhance security, improve performance, and simplify network management.

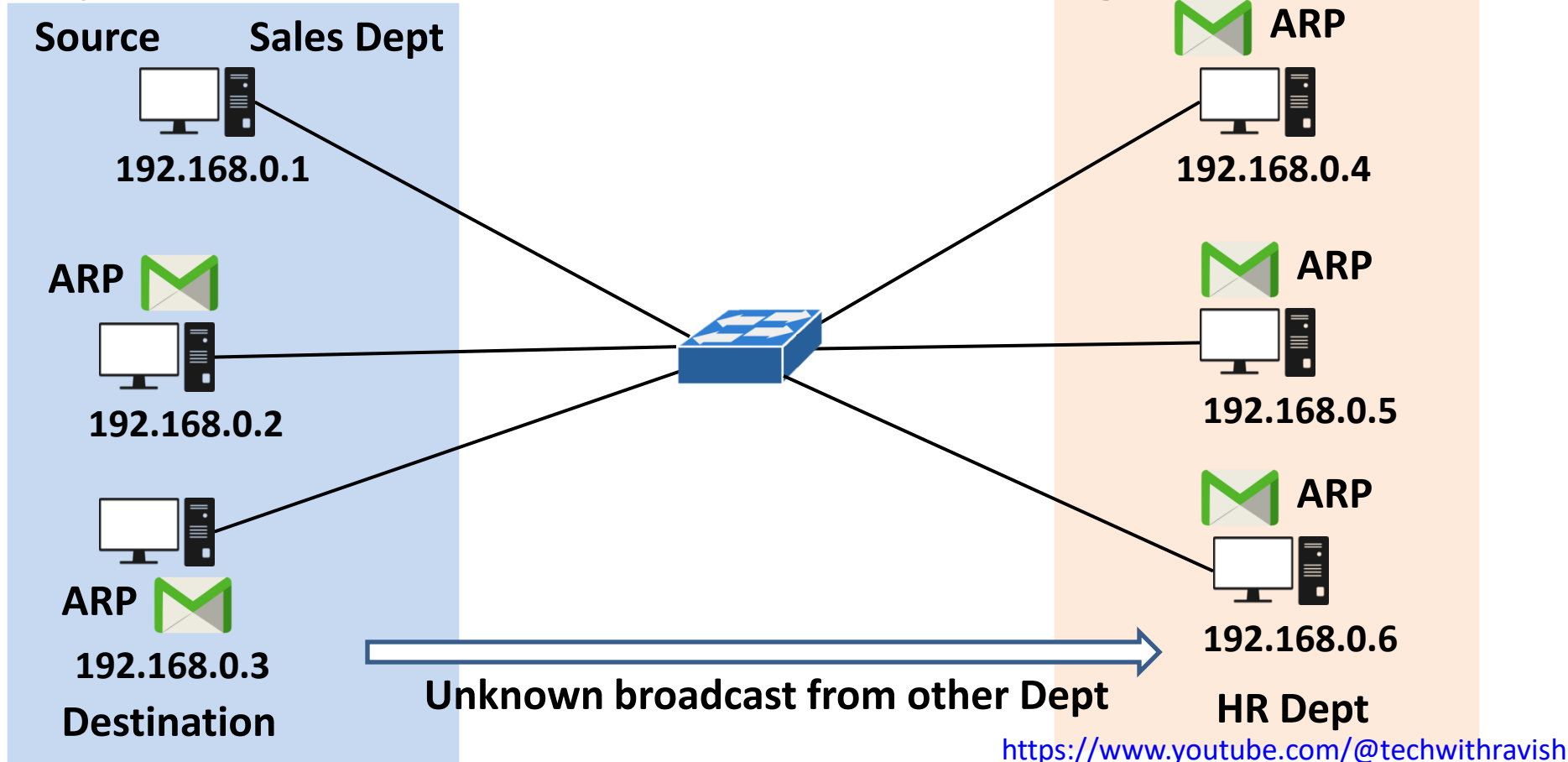
Key features and benefits of VLANs include:

1. Network Segmentation
2. Enhanced Security
3. Simplified Network Management
4. Improved Performance
5. Flexibility and Scalability

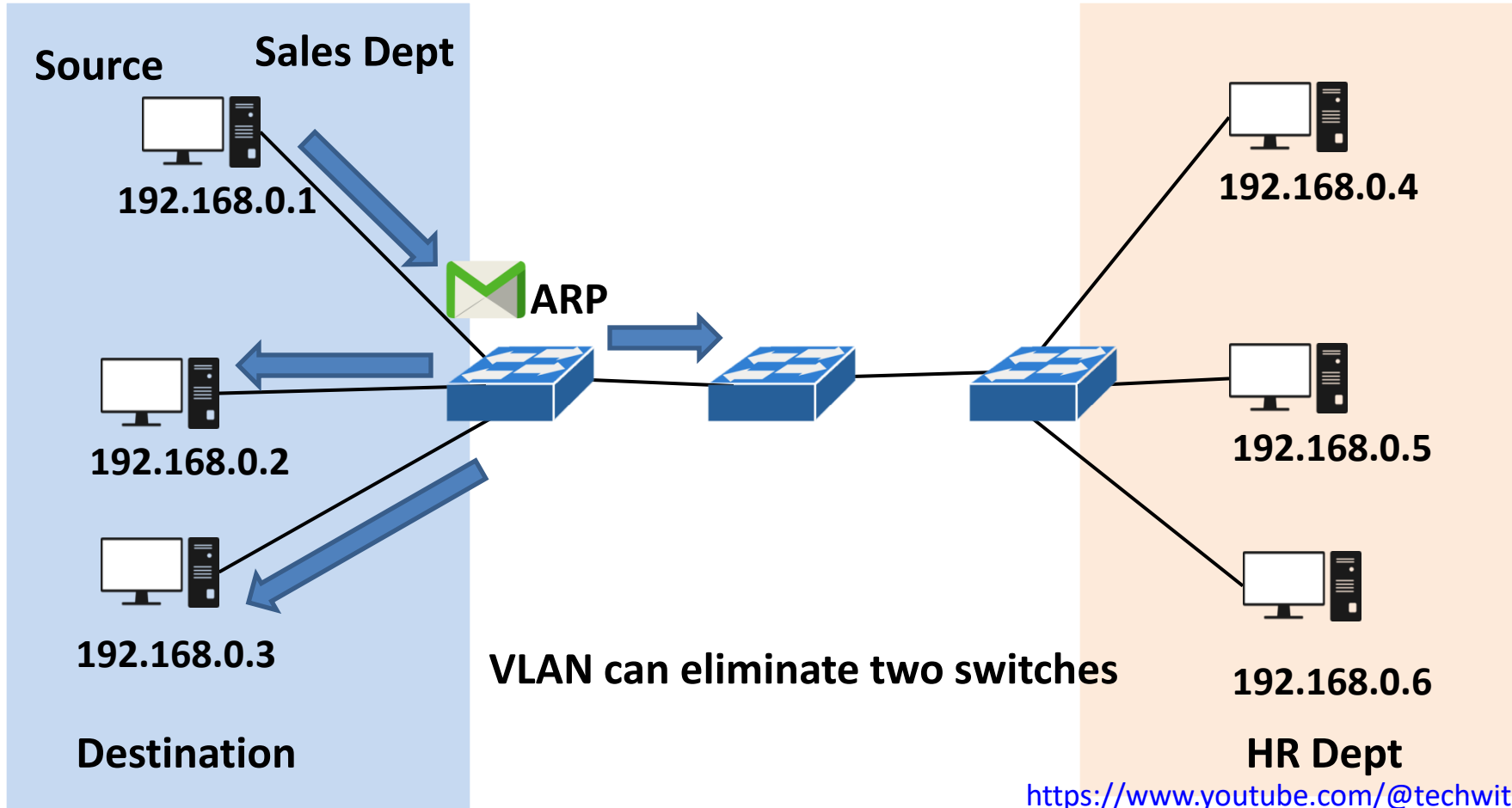
Issue: Without VLAN configuration the Broadcast from one department will be forwarded to another causing disturbance



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***Not a Cost Effective Solution to stop broadcast in other dept.**



VLAN: Virtual Local Area Network

- VLAN is Virtual local Area Network
- It is a logical network
- Support only Switches(L2 and L3)
- With the Help of VLAN we can break Single VLAN into Multiple VLAN
- Size 12 Bit
- No. of VLAN = $2^{12}=4096$
- Two Types of VLAN
 - > Ethernet VLAN
 - > Token Ring VLAN

Reserved VLAN Information:

VLAN No. 0 → By Default Assigned to Voice VLAN

VLAN No. 1 → By Default Available for All

(All ports are member of VLAN 1)

Can never Deleted/Created/Modified

Standard VLAN → Range 1-1001

Extended VLAN → Range 1006 – 4096

1002 VLAN → fddi-default active

1003 VLAN → token-ring-default active

1004 VLAN → fddinet-default active

1005 VLAN → trnet-default active

The VLANs 1002-1005 are still "unsupported"

VLAN Database:

Automatically Created when create any VLAN by admin

All VLAN Information Stored in VLAN Database

File Extension: VLAN.dat

Stored In: Flash Memory

Command to Check VLAN Database in Flash Memory:

Switch0#Show Flash:/

Switch0#Delete Flash:vlan.dat

VLAN Commands:

Create VLAN:

Switch(Config): VLAN _____

Switch(config-vlan)#Name _____

Create Range of VLAN:

Switch(Config): VLAN _____ - _____

Switch(Config): VLAN _____, _____, _____,

Delete VLAN:

Switch(Config): No VLAN _____

Switch(Config): No VLAN _____ - _____

Switch(Config): No VLAN _____, _____, _____,

Standard and Extended VLAN

- Standard VLAN is by default enabled on Switches
- Extended VLAN Manually created on Switches

How to create Extended VLAN:

-Extended VLAN Can only be Created in **Transparent mode VTP** in Switches

Create Extended VLAN:

Switch(Config): VTP Mode Transparent

Switch(Config): VLAN _____

Switch(config-vlan)#Name _____

Switch(config-vlan)#Exit

Alloting Interfaces to VLAN

Create VLAN:

Switch(Config): VLAN _____

Switch(config-vlan)#Name _____

Switch(config)#int fa0/_____

Switch(config-if)#switchport mode access

Switch(config-if)#switchport access vlan _____

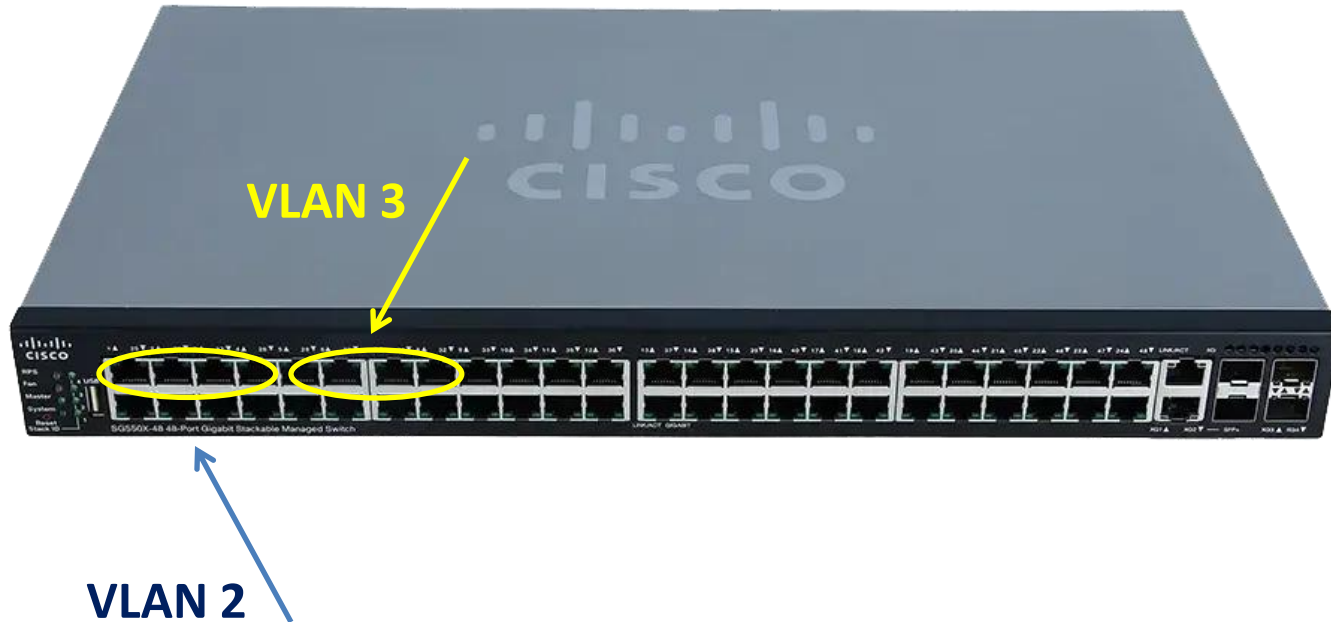
Create VLAN on Multiple Interfaces:

Switch(config)#int range fa0/_____-_____

Switch(config-if)#switchport mode access

Switch(config-if)#switchport access vlan _____

Virtual Segmentation of Switches:-



Example: Create VLAN on 3 interface of Switch

Create VLAN:

Switch(Config): VLAN 2

Switch(config-vlan)#Name Sales

Create VLAN on Multiple Interfaces:

Switch(config)#int range fa0/1-3

Switch(config-if)#switchport mode access

Switch(config-if)#switchport access vlan 2

